

# Fleet Performance & Delivery Efficiency Dashboard

## Technical Documentation

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**Tools Used:** Microsoft Excel | Power Query | Power BI Desktop | DAX

### 1. Project Overview

#### Introduction

The Fleet Performance & Delivery Efficiency Dashboard was developed to analyze fleet operations and monitor delivery performance using Microsoft Excel and Power BI. The project demonstrates an end-to-end Business Intelligence workflow, including data cleaning, transformation, data modelling, DAX calculations, AI-powered analytics, and interactive dashboard development.

The dashboard enables users to monitor fleet performance, evaluate fuel efficiency, compare maintenance costs, and identify operational factors affecting delivery performance.

### 2. Dataset Overview

#### Fact Table – Trip Data

| Attribute  | Description            |
|------------|------------------------|
| Trip_ID    | Unique trip identifier |
| Vehicle_ID | Vehicle identifier     |
| Driver_ID  | Driver identifier      |
| Origin     | Starting location      |

|                 |                          |
|-----------------|--------------------------|
| Destination     | Delivery destination     |
| Distance        | Distance travelled (km)  |
| Fuel Consumed   | Fuel consumed (Litres)   |
| Delivery Status | On-Time / Late           |
| Delivery Date   | Date of delivery         |
| Fuel Efficiency | Calculated column (km/L) |

**Dimension Table – Vehicle Master**

| Attribute        | Description               |
|------------------|---------------------------|
| Vehicle_ID       | Vehicle identifier        |
| Vehicle_Type     | Type of vehicle           |
| Capacity_kg      | Vehicle carrying capacity |
| Maintenance_Cost | Maintenance cost          |

### **3. Tools & Technologies**

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#### **Microsoft Power Query**

- **Data Import**
- **Data Cleaning**
- **Data Transformation**
- **Data Type Conversion**
- **Data Validation**

#### **Microsoft Power BI**

- **Data Modelling**
- **Relationship Management**
- **DAX Measures**
- **Interactive Dashboard**
- **AI Visuals**
- **Business Intelligence Reporting**

### **4. Data Pre-processing (Power Query)**

The dataset was prepared using Power Query before being loaded into Power BI.

#### **Tasks Performed**

- Imported datasets into Power Query.
- Removed duplicate records.
- Checked and handled missing or inconsistent values.
- Converted columns to appropriate data types.
- Standardized column names.
- Validated Vehicle IDs across tables.
- Loaded the transformed tables into Power BI.

### **5. Data Modelling**

A Star Schema data model was implemented to improve reporting performance and simplify data analysis.

#### Fact Table

Trip Data

#### Dimension Table

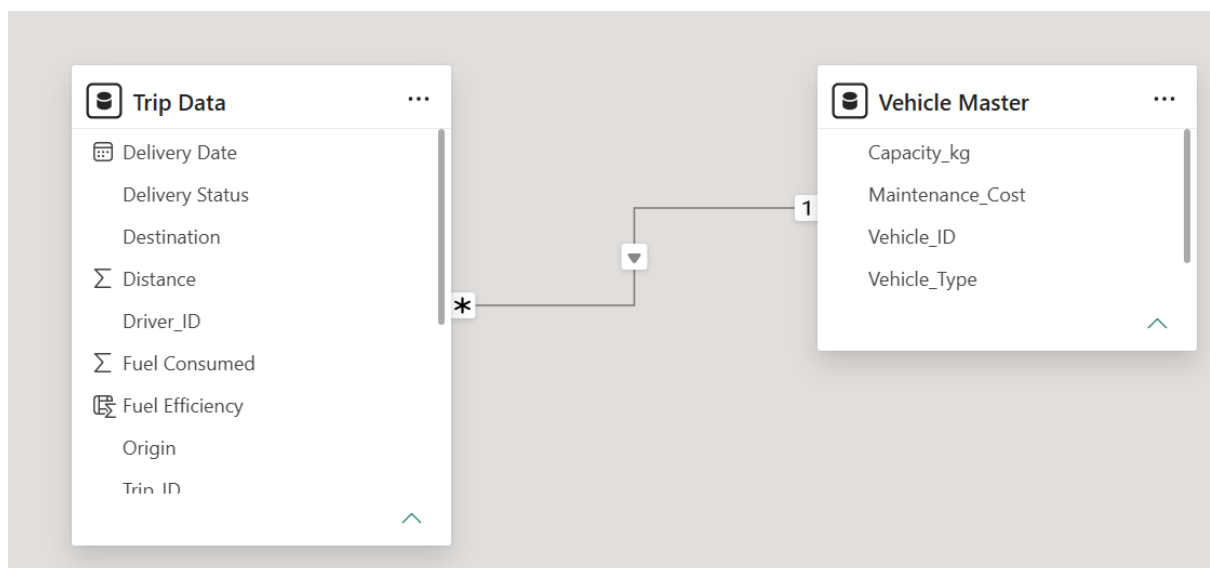
Vehicle Master

#### Relationship

| From Table | Column     | To Table       | Column     | Cardinality       |
|------------|------------|----------------|------------|-------------------|
| Trip Data  | Vehicle_ID | Vehicle Master | Vehicle_ID | Many-to-One (*:1) |

#### Data Model

Figure 1. Star Schema Data Model



## 6. Calculated Column

| Column          | Description                                                          |
|-----------------|----------------------------------------------------------------------|
| Fuel Efficiency | Calculates the distance travelled per litre of fuel consumed (km/L). |

## 7. DAX Measures

| Measure                 | Description                                      |
|-------------------------|--------------------------------------------------|
| Total Trips             | Counts the total number of trips.                |
| Total Fuel Consumed     | Calculates the total fuel consumed.              |
| On-Time Trips           | Counts deliveries completed on time.             |
| On-Time Delivery %      | Calculates the percentage of on-time deliveries. |
| Average Fuel Efficiency | Calculates average fuel efficiency (km/L).       |
| Cost per km             | Calculates transportation cost per kilometre.    |

|                          |                                                  |
|--------------------------|--------------------------------------------------|
| Average Maintenance Cost | Calculates average maintenance cost of vehicles. |
|--------------------------|--------------------------------------------------|

## 8. Dashboard

The Power BI dashboard provides an interactive overview of fleet operations.

### KPI Cards

- Total Trips
- Total Fuel Consumed
- On-Time Trips
- On-Time Delivery %
- Average Fuel Efficiency
- Cost per km

### Visualizations

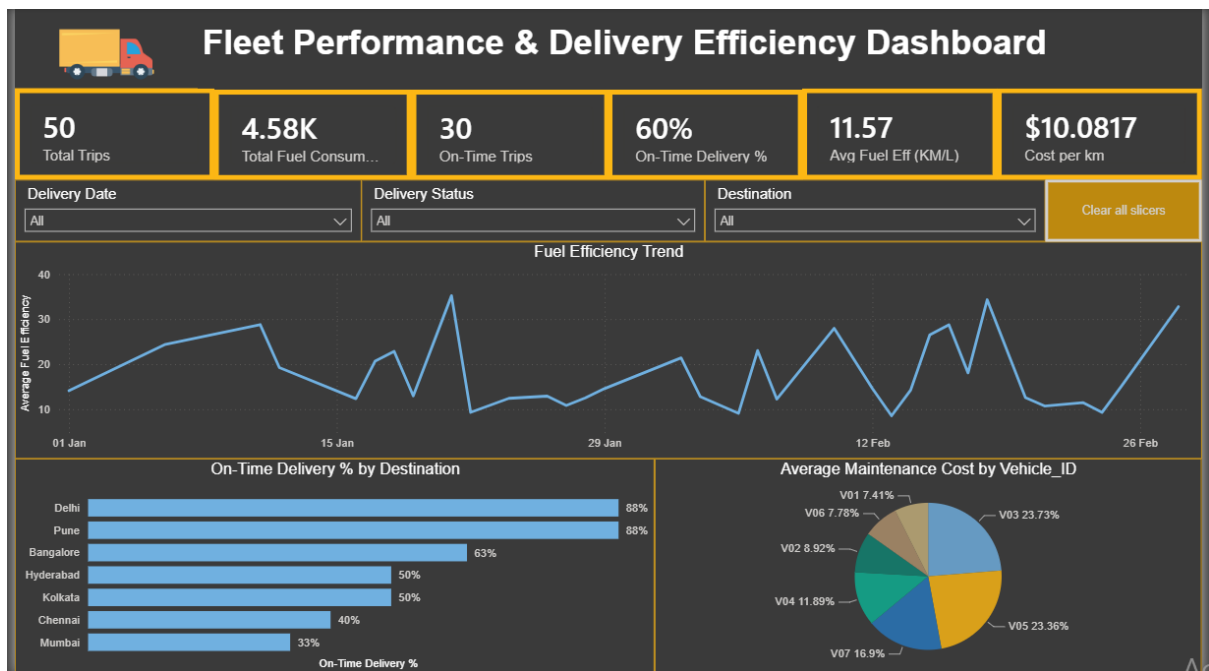
- Line Chart – Fuel Efficiency Trend
- Bar Chart – On-Time Delivery % by Destination
- Pie Chart – Average Maintenance Cost by Vehicle Type

### Interactive Features

- Delivery Date Slicer
- Delivery Status Slicer
- Destination Slicer
- Clear All Slicers Button

## Dashboard

**Figure 2. Fleet Performance Dashboard**



## 9. AI-Powered Analytics

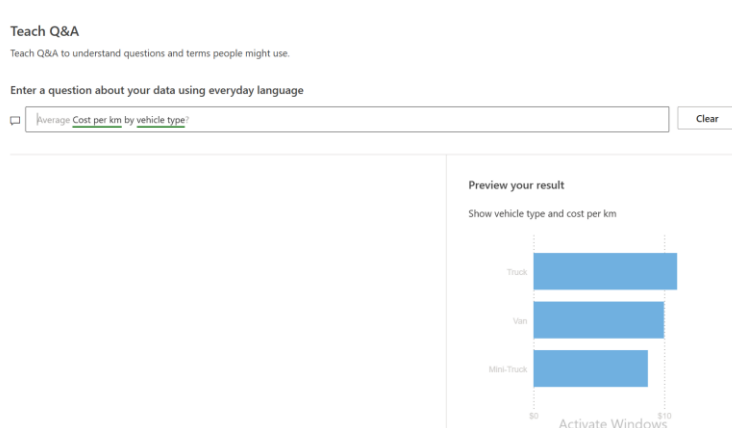
### 9.1 Natural Language Query (Teach Q&A)

#### Query Used

Average Cost per km by vehicle type

Power BI interpreted the natural language query and automatically generated the corresponding visualization.

**Figure 3. Teach Q&A**



### 9.2 Key Influencers

#### Analyze

- Delivery Status

### Explain By

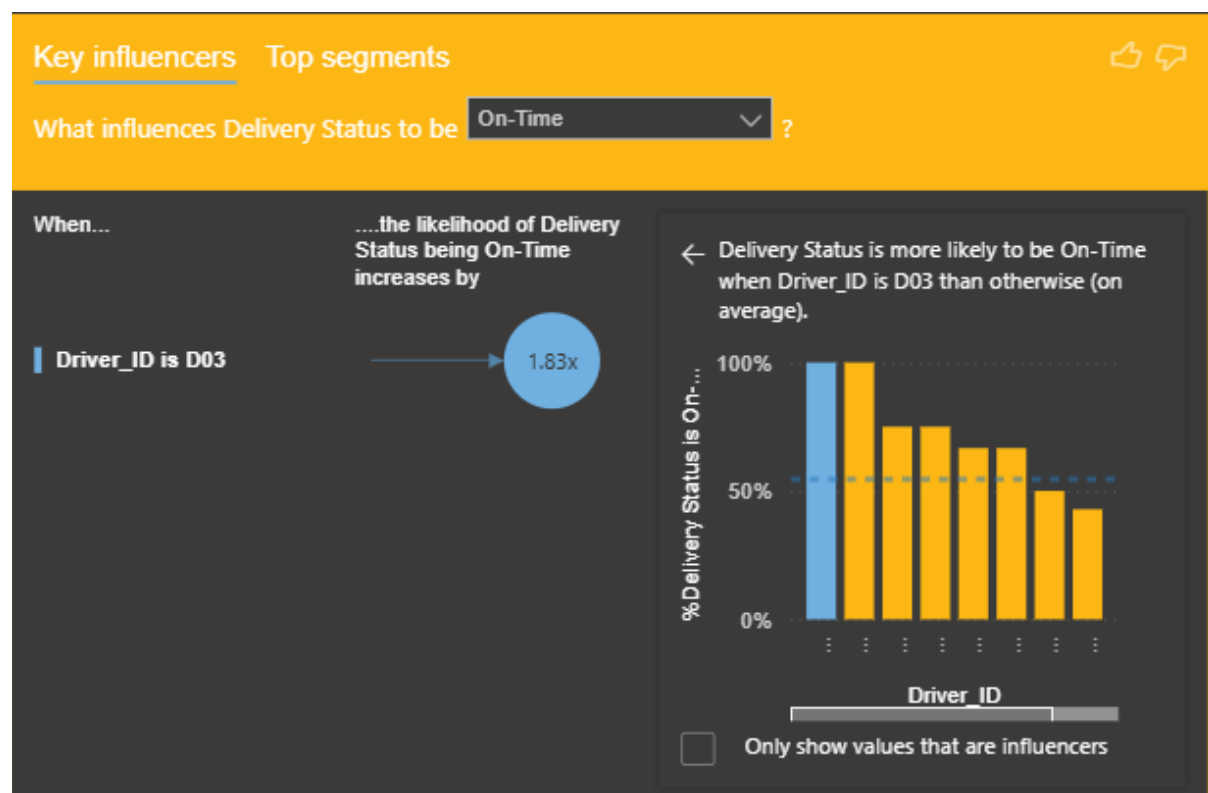
- Distance
- Vehicle\_Type
- Driver\_ID

### Insight

Power BI identified **Driver D03** as the strongest factor influencing on-time deliveries.

Trips completed by **Driver D03** were approximately **1.83× more likely** to be delivered on time than those handled by other drivers.

**Figure 4. Key Influencers**



## 9.3 Decomposition Tree

### Analyze

- Cost per km

### Explain By

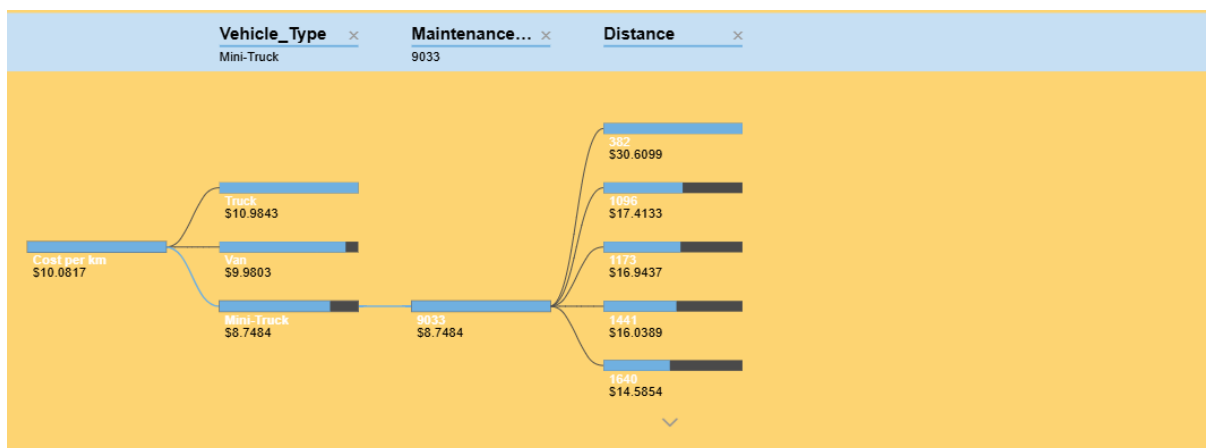
- Vehicle\_Type

- Maintenance\_Cost
- Distance

## Purpose

The decomposition tree enables users to drill down into transportation costs by vehicle type, maintenance cost, and trip distance, providing deeper insight into operational cost drivers.

**Figure 5. Decomposition Tree**



## 10. Key Insights

- 50 delivery trips were analyzed.
- 60% of deliveries were completed on time.
- The fleet consumed approximately **4.58K litres** of fuel.
- Average fuel efficiency was **11.57 km/L**.
- Average transportation cost was **₹10.08 per km**.
- Delhi and Pune recorded the highest on-time delivery performance.
- Truck vehicles incurred the highest maintenance costs.
- AI analysis identified **Driver D03** as the strongest factor associated with on-time deliveries.

## 11. Skills Demonstrated

- Data Import & Transformation

- Data Cleaning
- Power Query
- Star Schema Data Modelling
- Relationship Management
- DAX Measures
- Calculated Columns
- Interactive Dashboard Development
- AI-Powered Analytics
- Natural Language Query (Teach Q&A)
- Business Intelligence Reporting

## 12. Conclusion

This project demonstrates an end-to-end Business Intelligence solution using **Power Query** and **Power BI**. Through data transformation, star schema modelling, DAX calculations, AI-powered analytics, and interactive visualizations, the dashboard provides actionable insights into fleet operations, delivery performance, fuel efficiency, and transportation costs. It showcases practical Business Intelligence techniques applicable to logistics and fleet management.